Unit 5

Virtual Memory Management: Basic of Virtual Memory, Demand Paging, Page Replacement Algorithm: FIFO, LRU, Optimal

Disk Management: Disk Structure, Disk Scheduling– FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK

File System: File concepts, File Attributes, File Operations, File Types.

File Access Method: Sequential Access ,Direct Access

Virtual Memory in OS

Virtual Memory is a storage scheme that provides user an illusion of having a very big main memory. This is done by treating a part of secondary memory as the main memory.

In this scheme, User can load the bigger size processes than the available main memory by having the illusion that the memory is available to load the process.

Instead of loading one big process in the main memory, the Operating System loads the different parts of more than one process in the main memory.

By doing this, the degree of multiprogramming will be increased and therefore, the CPU utilization will also be increased.

How Virtual Memory Works?

In modern word, virtual memory has become quite common these days. In this scheme, whenever some pages needs to be loaded in the main memory for the execution and the memory is not available for those many pages, then in that case, instead of stopping the pages from entering in the main memory, the OS search for the RAM area that are least used in the recent times or that are not referenced and copy that into the secondary memory to make the space for the new pages in the main memory.

Since all this procedure happens automatically, therefore it makes the computer feel like it is having the unlimited RAM.

Demand Paging

Demand Paging is a popular method of virtual memory management. In demand paging, the pages of a process which are least used, get stored in the secondary memory.

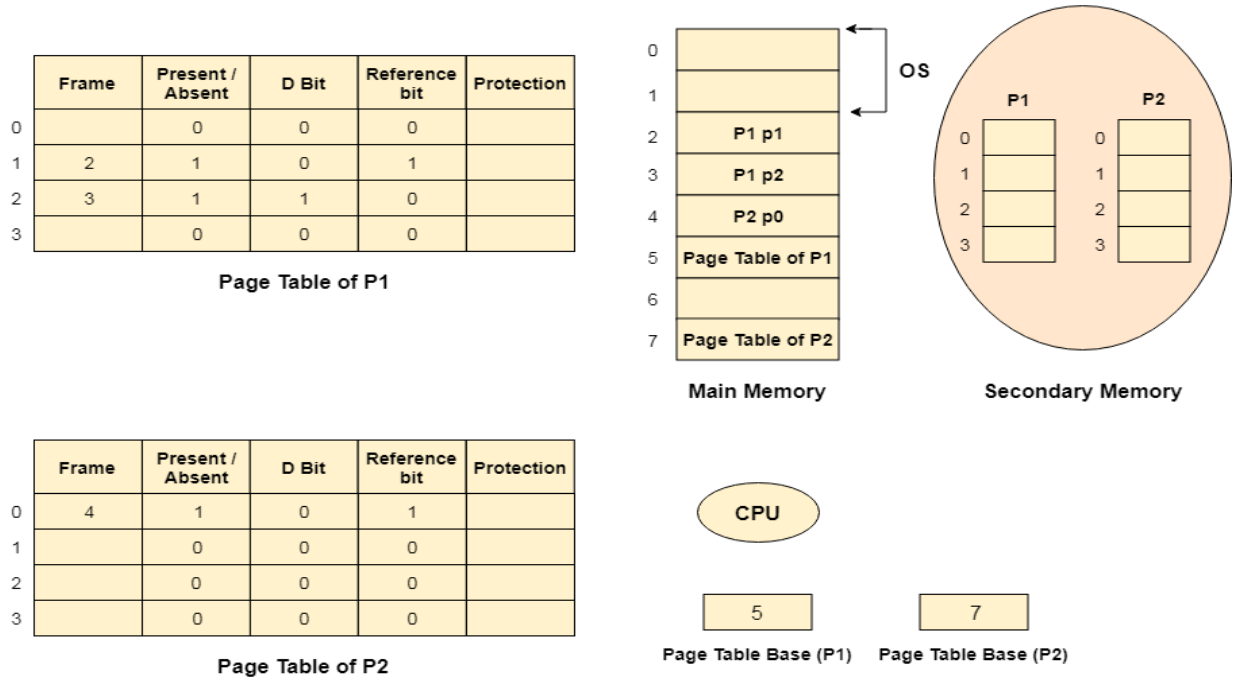
A page is copied to the main memory when its demand is made or page fault occurs. There are various page replacement algorithms which are used to determine the pages which will be replaced. We will discuss each one of them later in detail.

Snapshot of a virtual memory management system

Let us assume 2 processes, P1 and P2, contains 4 pages each. Each page size is 1 KB. The main memory contains 8 frame of 1 KB each. The OS resides in the first two partitions. In the third partition, 1st page of P1 is stored and the other frames are also shown as filled with the different pages of processes in the main memory.

The page tables of both the pages are 1 KB size each and therefore they can be fit in one frame each. The page tables of both the processes contain various information that is also shown in the image.

The CPU contains a register which contains the base address of page table that is 5 in the case of P1 and 7 in the case of P2. This page table base address will be added to the page number of the Logical address when it comes to accessing the actual corresponding entry.



Advantages of Virtual Memory

1. The degree of Multiprogramming will be increased.
2. User can run large application with less real RAM.
3. There is no need to buy more memory RAMs.

Disadvantages of Virtual Memory

1. The system becomes slower since swapping takes time.
2. It takes more time in switching between applications.
3. The user will have the lesser hard disk space for its use.

Demand Paging in OS

According to the concept of Virtual Memory, in order to execute some process, only a part of the process needs to be present in the main memory which means that only a few pages will only be present in the main memory at any time.

However, deciding, which pages need to be kept in the main memory and which need to be kept in the secondary memory, is going to be difficult because we cannot say in advance that a process will require a particular page at particular time.

Therefore, to overcome this problem, there is a concept called Demand Paging is introduced. It suggests keeping all pages of the frames in the secondary memory until they are required. In other words, it says that do not load any page in the main memory until it is required.

Whenever any page is referred for the first time in the main memory, then that page will be found in the secondary memory.

After that, it may or may not be present in the main memory depending upon the page replacement algorithm which will be covered later in this tutorial.

What is a Page Fault?

If the referred page is not present in the main memory then there will be a miss and the concept is called Page miss or page fault.

The CPU has to access the missed page from the secondary memory. If the number of page fault is very high then the effective access time of the system will become very high.

What is Thrashing?

If the number of page faults is equal to the number of referred pages or the number of page faults are so high so that the CPU remains busy in just reading the pages from the secondary memory then the effective access time will be the time taken by the CPU to read one word from the secondary memory and it will be so high. The concept is called thrashing.

If the page fault rate is PF %, the time taken in getting a page from the secondary memory and again restarting is S (service time) and the memory access time is ma then the effective access time can be given as;

1. EAT = PF X S + (1 - PF) X (ma)

Inverted Page Table

Inverted Page Table is the global page table which is maintained by the Operating System for all the processes. In inverted page table, the number of entries is equal to the number of frames in the main memory. It can be used to overcome the drawbacks of page table.

There is always a space reserved for the page regardless of the fact that whether it is present in the main memory or not. However, this is simply the wastage of the memory if the page is not present.

Pages	Frames
0	X
1	X
2	F1
3	F3
4	F6
5	X
6	F5

Page Table of P1

Pages	Frames
0	F2
1	F4
2	F7
3	X
4	X
5	X
6	F0

Page Table of P2

We can save this wastage by just inverting the page table. We can save the details only for the pages which are present in the main memory. Frames are the indices and the information saved inside the block will be Process ID and page number.

Pages	Frames
0	OS
1	P1 p2
2	P2 p0
3	P1 p3
4	P2 p1
5	P1 p6
6	P1 p4
7	P2 p2

Inverted Page Table

Page Replacement Algorithms in Operating Systems (OS)

There are three types of Page Replacement Algorithms. They are:

- Optimal Page Replacement Algorithm
- First In First Out Page Replacement Algorithm
- Least Recently Used (LRU) Page Replacement Algorithm

First in First Out Page Replacement Algorithm

This is the first basic algorithm of Page Replacement Algorithms. This algorithm is basically dependent on the number of frames used. Then each frame takes up the certain page and tries to access it. When the frames are filled then the actual problem starts. The fixed number of frames is filled up with the help of first frames present. This concept is fulfilled with the help of Demand Paging

After filling up of the frames, the next page in the waiting queue tries to enter the frame. If the frame is present then, no problem is occurred. Because of the page which is to be searched is already present in the allocated frames.

If the page to be searched is found among the frames then, this process is known as Page Hit.

If the page to be searched is not found among the frames then, this process is known as Page Fault.

When Page Fault occurs this problem arises, then the First In First Out Page Replacement Algorithm comes into picture.

The First In First Out (FIFO) Page Replacement Algorithm removes the Page in the frame which is allotted long back. This means the useless page which is in the frame for a longer time is removed and the new page which is in the ready queue and is ready to occupy the frame is allowed by the First In First Out Page Replacement.

Let us understand this First In First Out Page Replacement Algorithm working with the help of an example.

Example:

Consider the reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0 for a memory with three frames and calculate number of page faults by using FIFO (First In First Out) Page replacement algorithms.

Points to Remember

Page Not Found - - - > Page Fault

Page Found - - - > Page Hit

Reference String:

6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0

S. no	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
F3				2	2	2	4	4	4	2	2	2	2	2	2	2	2	2	2	0
F2		1	1	1	1	3	3	3	0	0	0	0	0	0	0	3	3	3	3	3
F1	6	6	6	6	0	0	0	6	6	6	1	1	1	1	1	1	1	1	1	1
Hit (H)/ Fault (F)	F	F	H	F	F	F	F	F	F	F	F	H	H	H	H	F	H	H	H	F

Number of Page Hits = 8
Number of Page Faults = 12
The Ratio of Page Hit to the Page Fault = 8 : 12 - - - > 2 : 3 - - - > 0.66
The Page Hit Percentage = 8 *100 / 20 = 40%
The Page Fault Percentage = 100 - Page Hit Percentage = 100 - 40 = 60%

Explanation

First, fill the frames with the initial pages. Then, after the frames are filled we need to create a space in the frames for the new page to occupy. So, with the help of First in First Out Page Replacement Algorithm we remove the frame which contains the page is older among the pages. By removing the older page, we give access for the new frame to occupy the empty space created by the First in First out Page Replacement Algorithm.

OPTIMAL Page Replacement Algorithm

This is the second basic algorithm of Page Replacement Algorithms. This algorithm is basically dependent on the number of frames used. Then each frame takes up the certain page and tries to access it. When the frames are filled then the actual problem starts. The fixed number of frames is filled up with the help of first frames present. This concept is fulfilled with the help of Demand Paging

After filling up of the frames, the next page in the waiting queue tries to enter the frame. If the frame is present then, no problem is occurred. Because of the page which is to be searched is already present in the allocated frames.

If the page to be searched is found among the frames then, this process is known as Page Hit.

If the page to be searched is not found among the frames then, this process is known as Page Fault.

When Page Fault occurs this problem arises, then the OPTIMAL Page Replacement Algorithm comes into picture.

The OPTIMAL Page Replacement Algorithms works on a certain principle. The principle is:

Replace the Page which is not used in the Longest Dimension of time in future

This principle means that after all the frames are filled then, see the future pages which are to occupy the frames. Go on checking for the pages which are already available in the frames. Choose the page which is at last.

Example:

Suppose the Reference String is:

0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0

6, 1, 2 are in the frames occupying the frames.

Now we need to enter 0 into the frame by removing one page from the page

So, let us check which page number occurs last

From the sub sequence 0, 3, 4, 6, 0, 2, 1 we can say that 1 is the last occurring page number. So we can say that 0 can be placed in the frame body by removing 1 from the frame.

Let us understand this OPTIMAL Page Replacement Algorithm working with the help of an example.

Example:

Consider the reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 4, 0 for a memory with three frames and calculate number of page faults by using OPTIMAL Page replacement algorithms.

Points to Remember

Page Not Found - - - > Page Fault

Page Found - - - > Page Hit

Reference String:

6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 4, 0

S. no	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
F3				2	2	2	4	4	4	4	4	4	4	4	4	3	3	3	4	4
F2		1	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
F1	6	6	6	6	0	0	0	6	6	2	2	2	2	2	2	2	2	2	2	0
Hit (H)/ Faul t (F)	F	F	H	F	F	F	F	H	H	F	F	H	H	H	H	F	H	F	F	F

Number of Page Hits = 8
Number of Page Faults = 12
The Ratio of Page Hit to the Page Fault = 8 : 12 - - - > 2 : 3 - - - > 0.66
The Page Hit Percentage = 8 *100 / 20 = 40%
The Page Fault Percentage = 100 - Page Hit Percentage = 100 - 40 = 60%

Explanation

First, fill the frames with the initial pages. Then, after the frames are filled we need to create a space in the frames for the new page to occupy.

Here, we would fill the empty spaces with the pages we and the empty frames we have. The problem occurs when there is no space for occupying of pages. We have already known that we would replace the Page which is not used in the Longest Dimension of time in future.

There comes a question what if there is absence of page which is in the frame.

Suppose the Reference String is:

0, 2, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0

6, 1, 5 are in the frames occupying the frames.

Here, we can see that page number 5 is not present in the Reference String. But the number 5 is present in the Frame. So, as the page number 5 is absent we remove it when required and other page can occupy that position.

Least Recently Used (LRU) Replacement Algorithm

This is the last basic algorithm of Page Replacement Algorithms. This algorithm is basically dependent on the number of frames used. Then each frame takes up the certain page and tries to access it. When the frames are filled then the actual problem starts. The fixed number of frames is filled up with the help of first frames present. This concept is fulfilled with the help of Demand Paging

After filling up of the frames, the next page in the waiting queue tries to enter the frame. If the frame is present then, no problem is occurred. Because of the page which is to be searched is already present in the allocated frames.

If the page to be searched is found among the frames then, this process is known as Page Hit.

If the page to be searched is not found among the frames then, this process is known as Page Fault.

When Page Fault occurs this problem arises, then the Least Recently Used (LRU) Page Replacement Algorithm comes into picture.

The Least Recently Used (LRU) Page Replacement Algorithms works on a certain principle. The principle is:

Replace the page with the page which is less dimension of time recently used page in the past.

Example:

Suppose the Reference String is:

6, 1, 1, 2, 0, 3, 4, 6, 0

The pages with page numbers 6, 1, 2 are in the frames occupying the frames.

Now, we need to allot a space for the page numbered 0.

Now, we need to travel back into the past to check which page can be replaced.

6 is the oldest page which is available in the Frame.

So, replace 6 with the page numbered 0.

Let us understand this Least Recently Used (LRU) Page Replacement Algorithm working with the help of an example.

Example:

Consider the reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0 for a memory with three frames and calculate number of page faults by using Least Recently Used (LRU) Page replacement algorithms.

Points to Remember

Page Not Found - - - > Page Fault

Page Found - - - > Page Hit

Reference String:

6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0

S. no	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
F3				2	2	2	4	4	4	2	2	2	2	2	2	2	2	2	2	2
F2		1	1	1	1	3	3	3	0	0	0	0	0	0	0	0	0	1	1	1
F1	6	6	6	6	0	0	0	6	6	6	1	1	1	1	1	3	3	3	3	0
Hit (H)/ Fault (F)	F	F	H	F	F	F	F	F	F	F	F	H	H	H	H	F	H	F	H	F

Number of Page Hits = 7
Number of Page Faults = 13
The Ratio of Page Hit to the Page Fault = 7 : 12 - - - > 0.5833 : 1
The Page Hit Percentage = 7 * 100 / 20 = 35%
The Page Fault Percentage = 100 - Page Hit Percentage = 100 - 35 = 65%

Explanation

First, fill the frames with the initial pages. Then, after the frames are filled we need to create a space in the frames for the new page to occupy.

Here, we would fill the empty spaces with the pages we and the empty frames we have. The problem occurs when there is no space for occupying of pages. We have already known that we would replace the Page which is not used in the Longest Dimension of time in past or can be said as the Page which is very far away in the past.

Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. the number of frames in the memory is 3. Find out the number of page faults respective to:

1. Optimal Page Replacement Algorithm
2. FIFO Page Replacement Algorithm
3. LRU Page Replacement Algorithm

Optimal Page Replacement Algorithm

Request	4	7	6	1	7	6	1	2	7	2
Frame 3			6	6	6	6	6	2	2	2
Frame 2		7	7	7	7	7	7	7	7	7
Frame 1	4	4	4	1	1	1	1	1	1	1
Miss/Hit	Miss	Miss	Miss	Miss	Hit	Hit	Hit	Miss	Hit	Hit

Number of Page Faults in Optimal Page Replacement Algorithm = 5

LRU Page Replacement Algorithm

Request	**4**	**7**	**6**	**1**	**7**	**6**	**1**	**2**	**7**	**2**
Frame 3			6	6	6	6	6	6	7	7
Frame 2		7	7	7	7	7	7	2	2	2
Frame 1	4	4	4	1	1	1	1	1	1	1
Miss/Hit	**Miss**	**Miss**	**Miss**	**Miss**	**Hit**	**Hit**	**Hit**	**Miss**	**Miss**	**Hit**

Number of Page Faults in LRU = 6

FIFO Page Replacement Algorithm

Request	**4**	**7**	**6**	**1**	**7**	**6**	**1**	**2**	**7**	**2**
Frame 3			6	6	6	6	6	6	7	7
Frame 2		7	7	7	7	7	7	2	2	2
Frame 1	4	4	4	1	1	1	1	1	1	1
Miss/Hit	Miss	Miss	Miss	Miss	Hit	Hit	Hit	Miss	Miss	Hit

Number of Page Faults in FIFO = 6

Disk Structure in OS

Hard Disk is a secondary storage device. It is a type of electro mechanical device. A hard disk stores and retrieves digital data using magnetic storage. Disk is rigid rapidly rotating platters coated with magnetic material.

Hard Disk Pack consist of more than one disk.

How Data is Stored in a Hard Disk

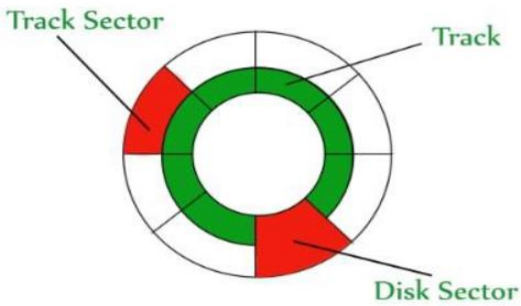
● Data is stored in Disk in form of tracks and sectors.

● Disk surface is divided in to tracks.

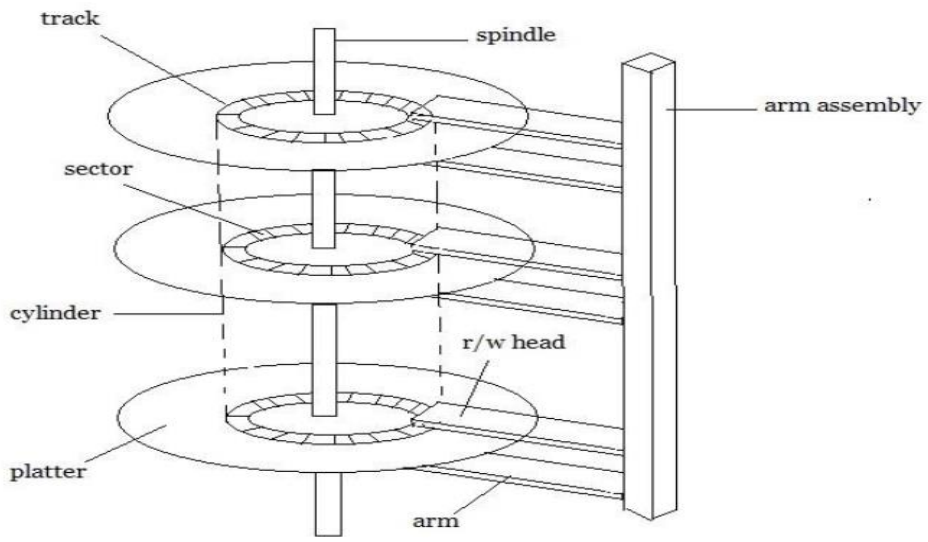
● Track is further dived in to sectors.

● Sector is the addressable unit in disk.

Basic picture for disk storage concept is given below



Disk structure is as shown in following figure



Step wise description of Disk Structure is given below

● Disk surface divided into tracks

● A read/write head positioned just above the disk surface

● Information stored by magnetic recording on the track under read/write head

● Fixed head disk

● Moving head disk

● Designed for large amount of storage

● Primary design consideration cost, size, and speed

Hardware for disk system

● Disk drive, Device motor, Read/write head, Associated logic

Disk controller

● Determines the logical interaction with the computer

● Can service more than one drive (overlapped seeks)

Cylinder

● The same numbered tracks on all the disk surfaces

● Each track contains between 8 to 32 sectors

Sector

● Smallest unit of information that can be read from/written into disk

● Range from 32 bytes to 4096 bytes

Disk Performance Parameters

Some important parameters used to measure the performance of hard disk are as follow

Seek time

● Seek Time is time required by read/write head to move to requested track

● Includes the initial startup time and the time required to traverse the tracks to be crossed once the access arm is up to speed.

Latency or rotational delay

● Time required for the requested sector to come under the read/write head.

● Rotational delay is generally the half of the time taken in one rotation.

Data Transfer Rate

Data transfer rate is define as the amount of data transfer in per unit time for example 30 MB/Sec.

Data Transfer Time

Data Transfer time is the total time taken to transfer a specific amount of data from the disk. Data Transfer time depends on the data transfer rate of the disk.

Average Access Time

Average access time is calculated as

Average Access Time = Seek Time + Rotational Latecny + Data Transfer Time

Disk Scheduling Algorithms

The list of various disks scheduling algorithm is given below. Each algorithm is carrying some advantages and disadvantages. The limitation of each algorithm leads to the evolution of a new algorithm.

- FCFS scheduling algorithm
- SSTF (shortest seek time first) algorithm
- SCAN scheduling
- C-SCAN scheduling
- LOOK Scheduling
- C-LOOK scheduling

FCFS Scheduling Algorithm

It is the simplest Disk Scheduling algorithm. It services the IO requests in the order in which they arrive. There is no starvation in this algorithm, every request is serviced.

Disadvantages

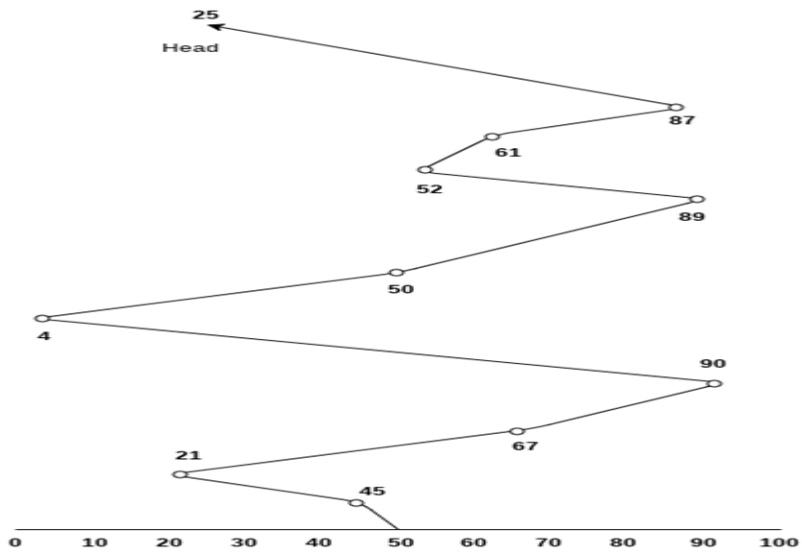
- The scheme does not optimize the seek time.
- The request may come from different processes therefore there is the possibility of inappropriate movement of the head.

Example

Consider the following disk request sequence for a disk with 100 tracks 45, 21, 67, 90, 4, 50, 89, 52, 61, 87, 25

Head pointer starting at 50 and moving in left direction. Find the number of head movements in cylinders using FCFS scheduling.

Solution



Number of cylinders moved by the head

= (50-45)+(45-21)+(67-21)+(90-67)+(90-4)+(50-4)+(89-50)+(61-52)+(87-61)+(87-25)

= 5 + 24 + 46 + 23 + 86 + 46 + 49 + 9 + 26 + 62

= 376

SSTF Scheduling Algorithm

Shortest seek time first (SSTF) algorithm selects the disk I/O request which requires the least disk arm movement from its current position regardless of the direction. It reduces the total seek time as compared to FCFS.

It allows the head to move to the closest track in the service queue.

Disadvantages

- It may cause starvation for some requests.
- Switching direction on the frequent basis slows the working of algorithm.
- It is not the most optimal algorithm.

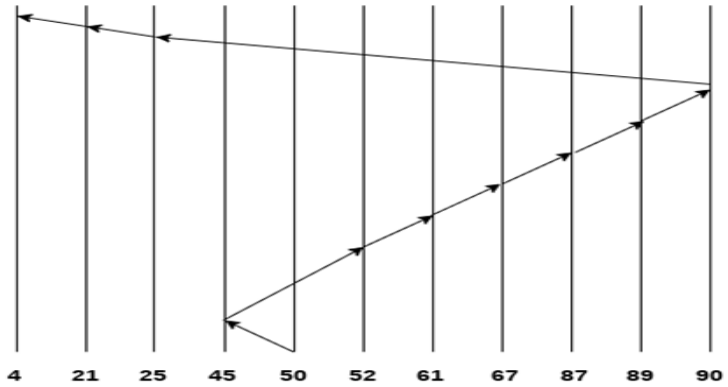
Example

Consider the following disk request sequence for a disk with 100 tracks

45, 21, 67, 90, 4, 89, 52, 61, 87, 25

Head pointer starting at 50. Find the number of head movements in cylinders using SSTF scheduling.

Solution:



Number of cylinders = 5 + 7 + 9 + 6 + 20 + 2 + 1 + 65 + 4 + 17 = 136

SCAN and C-SCAN algorithm

Scan Algorithm

It is also called as Elevator Algorithm. In this algorithm, the disk arm moves into a particular direction till the end, satisfying all the requests coming in its path,and then it turns backand moves in the reverse direction satisfying requests coming in its path.

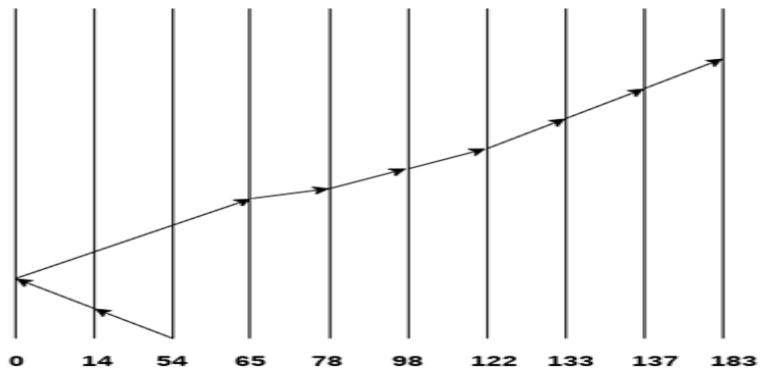
It works in the way an elevator works, elevator moves in a direction completely till the last floor of that direction and then turns back.

Example

Consider the following disk request sequence for a disk with 100 tracks

98, 137, 122, 183, 14, 133, 65, 78

Head pointer starting at 54 and moving in left direction. Find the number of head movements in cylinders using SCAN scheduling.



Number of Cylinders = 40 + 14 + 65 + 13 + 20 + 24 + 11 + 4 + 46 = 237

C-SCAN algorithm

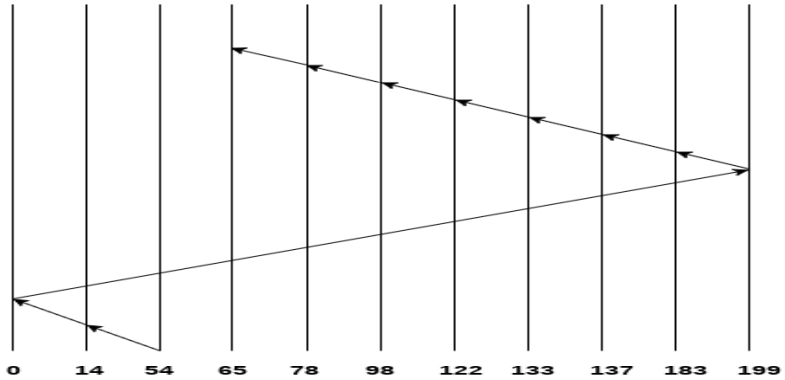
In C-SCAN algorithm, the arm of the disk moves in a particular direction servicing requests until it reaches the last cylinder, then it jumps to the last cylinder of the opposite direction without servicing any request then it turns back and start moving in that direction servicing the remaining requests.

Example

Consider the following disk request sequence for a disk with 100 tracks

98, 137, 122, 183, 14, 133, 65, 78

Head pointer starting at 54 and moving in left direction. Find the number of head movements in cylinders using C-SCAN scheduling.



No. of cylinders crossed = 40 + 14 + 199 + 16 + 46 + 4 + 11 + 24 + 20 + 13 = 387

Look Scheduling

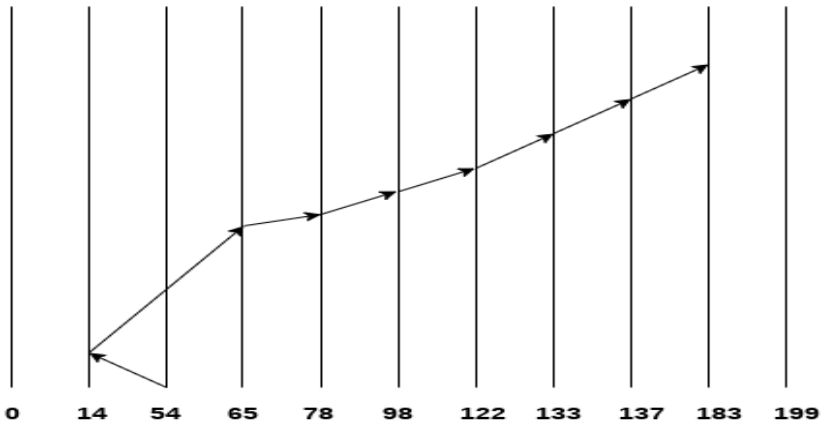
It is like SCAN scheduling Algorithm to some extant except the difference that, in this scheduling algorithm, the arm of the disk stops moving inwards (or outwards) when no more request in that direction exists. This algorithm tries to overcome the overhead of SCAN algorithm which forces disk arm to move in one direction till the end regardless of knowing if any request exists in the direction or not.

Example

Consider the following disk request sequence for a disk with 100 tracks

98, 137, 122, 183, 14, 133, 65, 78

Head pointer starting at 54 and moving in left direction. Find the number of head movements in cylinders using LOOK scheduling.



Number of cylinders crossed = 40 + 51 + 13 + +20 + 24 + 11 + 4 + 46 = 209

C Look Scheduling

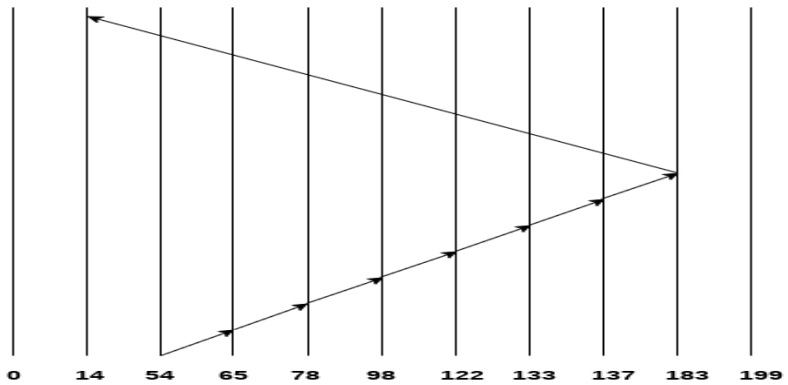
C Look Algorithm is similar to C-SCAN algorithm to some extent. In this algorithm, the arm of the disk moves outwards servicing requests until it reaches the highest request cylinder, then it jumps to the lowest request cylinder without servicing any request then it again start moving outwards servicing the remaining requests.

It is different from C SCAN algorithm in the sense that, C SCAN force the disk arm to move till the last cylinder regardless of knowing whether any request is to be serviced on that cylinder or not.

Example

Consider the following disk request sequence for a disk with 100 tracks 98, 137, 122, 183, 14, 133, 65, 78

Head pointer starting at 54 and moving in left direction. Find the number of head movements in cylinders using C LOOK scheduling.

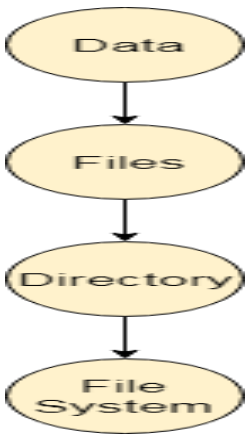


Number of cylinders crossed = 11 + 13 + 20 + 24 + 11 + 4 + 46 + 169 = 298

File systems Management: File concept, Access methods

A file can be defined as a data structure which stores the sequence of records. Files are stored in a file system, which may exist on a disk or in the main memory. Files can be simple (plain text) or complex (specially-formatted).

The collection of files is known as Directory. The collection of directories at the different levels, is known as File System.



Attributes of the File

1.Name

Every file carries a name by which the file is recognized in the file system. One directory cannot have two files with the same name.

2.Identifier

Along with the name, Each File has its own extension which identifies the type of the file. For example, a text file has the extension **.txt,** A video file can have the extension **.mp4.**

3.Type

In a File System, the Files are classified in different types such as video files, audio files, text files, executable files, etc.

4.Location

In the File System, there are several locations on which, the files can be stored. Each file carries its location as its attribute.

5.Size

The Size of the File is one of its most important attribute. By size of the file, we mean the number of bytes acquired by the file in the memory.

6.Protection

The Admin of the computer may want the different protections for the different files. Therefore, each file carries its own set of permissions to the different group of Users.

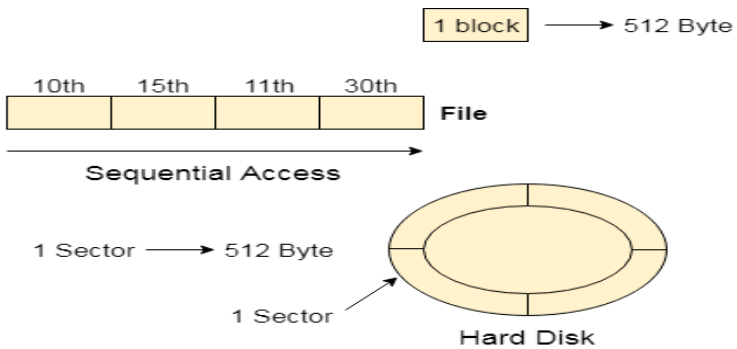
7.Time and Date

Every file carries a time stamp which contains the time and date on which the file is last modified.

File Access Methods

Let's look at various ways to access files stored in secondary memory.

Sequential Access



Most of the operating systems access the file sequentially. In other words, we can say that most of the files need to be accessed sequentially by the operating system.

In sequential access, the OS read the file word by word. A pointer is maintained which initially points to the base address of the file. If the user wants to read first word of the file then the pointer provides that word to the user and increases its value by 1 word. This process continues till the end of the file.

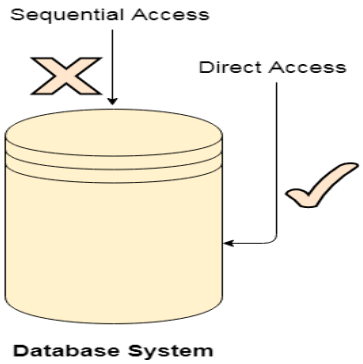
Modern word systems do provide the concept of direct access and indexed access but the most used method is sequential access due to the fact that most of the files such as text files, audio files, video files, etc need to be sequentially accessed.

Direct Access

The Direct Access is mostly required in the case of database systems. In most of the cases, we need filtered information from the database. The sequential access can be very slow and inefficient in such cases.

Suppose every block of the storage stores 4 records and we know that the record we needed is stored in 10th block. In that case, the sequential access will not be implemented because it will traverse all the blocks in order to access the needed record.

Direct access will give the required result despite of the fact that the operating system has to perform some complex tasks such as determining the desired block number. However, that is generally implemented in database applications.



Indexed Access

If a file can be sorted on any of the filed then an index can be assigned to a group of certain records. However, A particular record can be accessed by its index. The index is nothing but the address of a record in the file.

In index accessing, searching in a large database became very quick and easy but we need to have some extra space in the memory to store the index value.